

Erdős Problem #84

EP-84

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

The cycle set of a graph G on n vertices is a set $A \subseteq \{3, \dots, n\}$ such that there is a cycle in G of length ℓ if and only if $\ell \in A$. Let $f(n)$ count the number of possible such A .

Prove that $f(n) = o(2^n)$.

Prove that $f(n)/2^{n/2} \rightarrow \infty$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>